



Jonathan Huecker
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RA-25-0188

10 CFR 50.73

July 21, 2025

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555

Subject: Duke Energy Carolinas, LLC
Catawba Nuclear Station, Units 1 and 2
Docket No. 50-413, 50-414
Licensee Event Report (LER) 2025-002-00

Pursuant to 10 CFR 50.73(a)(1) and (d), attached is LER 413/2025-002-00, entitled "Inoperability of 1B Emergency Diesel Generator due to Fuel Oil Leak". This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B).

There are no regulatory commitments contained in this letter or its attachment.

The impact of this event in relation to plant risk was low and there was no impact to the health and safety of the public.

If questions arise regarding this LER, please contact Ari Tuckman of Regulatory Affairs at (803) 701-3771.

Sincerely,

A handwritten signature in black ink, appearing to read "J. Huecker", written over a horizontal line.

Jonathan Huecker
Vice President, Catawba Nuclear Station

Attachment

United States Nuclear Regulatory Commission
Page 2
July 21, 2025

xc (with attachment):

Julio Lara
Acting Regional Administrator
U.S. Nuclear Regulatory Commission - Region II

Jack Minzer-Bryant
NRC Project Manager (CNS)
U.S. Nuclear Regulatory Commission

David Rivard
Senior Resident Inspector, Catawba Nuclear Station
U.S. Nuclear Regulatory Commission



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Catawba Nuclear Station, Unit 1	☒ 050 ☐ 052	2. Docket Number 00413	3. Page 1 of 5
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4. Title
Inoperability of 1B Emergency Diesel Generator due to Fuel Oil Leak

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 052	Docket Number
04	23	2025	2025	002	00	07	21	25	Catawba Unit 2	☒	00414
									Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Ari Tuckman, Lead Engineer, CNS Regulatory Affairs	Phone Number (Include area code) (803) 701-3771
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	DC	TBG	NA	Y					

14. Supplemental Report Expected

15. Expected Submission Date

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 4/23/2025, Catawba Unit 1 was in Mode 1 operating at approximately 100 percent power. At 0248 hours, the 1B Emergency Diesel Generator (EDG) was started for a 24 hour surveillance. Approximately 11 hours into the surveillance a leak originated on the common fuel supply line under a compression fitting. The leak was estimated to be greater than 0.5 gpm and at 1457 hours the 1B EDG was secured and declared inoperable at 1459 hours.

Repairs were made and the 1B EDG was returned to service on 4/25/2025 at 0600 hours. Investigation revealed a circumferential crack located within a tubing fitting. The crack originated prior to the attempted 24 hour surveillance but was not through-wall and there was no indication of leakage. The crack propagated during the surveillance due to high cycle fatigue. The existence of the crack, even though no leakage was present, prior to the 24-hour surveillance is evidence that the 1B EDG was inoperable for a time longer than allowed by Technical Specifications.

This event is being reported pursuant to 10 CFR 50.73(a)(2)(i)(B) as operation or condition prohibited by Technical Specifications. There was no impact to the health and safety of the public.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME		☒ 050	2. DOCKET NUMBER		3. LER NUMBER		
Catawba Nuclear Station, Units 1 and 2		☐ 052	00413, 00414		YEAR	SEQUENTIAL NUMBER	REV NO.
					2025	002	00

NARRATIVE

BACKGROUND

The following information is provided to assist readers in understanding the event described in this LER. Applicable Energy Industry Identification System [EII] and component codes are enclosed within brackets. Catawba's unique system and component identifiers are contained within parentheses.

The onsite standby power source for each 4160 volt Engineered Safety Features (ESF) bus at Catawba is a dedicated EDG. For each unit, EDGs A and B are dedicated to ESF buses ETA and ETB, respectively. Each EDG starts automatically on a Safety Injection signal (i.e., low pressurizer pressure or high containment pressure) or on an ESF bus degraded voltage or undervoltage signal. After the EDG has started, it will automatically tie to its respective bus after offsite power is tripped as a consequence of ESF bus undervoltage or degraded voltage, independent of or coincident with an SI signal. The EDGs will also start and operate in the standby mode without tying to the ESF bus on an SI signal alone. Following the trip of offsite power, a sequencer strips loads from the ESF bus. When the EDG is tied to the ESF bus, loads are then sequentially connected to its respective ESF bus by the automatic load sequencer. In the event of a loss of preferred power, the ESF electrical loads are automatically connected to the EDGs in sufficient time to provide for safe reactor shutdown and to mitigate the consequences of a Design Basis Accident such as a Loss of Coolant Accident (LOCA).

There are also provisions to accommodate the connecting of the Emergency Supplemental Power Source (ESPS) to one train of either unit's Class 1E AC Distribution System. The ESPS consists of two 50% capacity non-safety related commercial grade EDGs. Manual actions are required to align the ESPS to the station and only one of the station's four onsite Class 1E Distribution System trains can be supplied by the ESPS at any given time. The ESPS is made available to support extended Completion Times in the event of an inoperable EDG as well as a defense-in-depth source of AC power to mitigate a station blackout event. The ESPS remains disconnected from the Class 1E AC Distribution System unless required for supplemental power to one of the four 4.16 kV ESF buses.

The primary purpose of the Diesel Fuel Oil System (FD) [DC] is to store and supply fuel to the diesel generators' engines for a period of operation of no less than seven days. Each diesel is equipped with two 45,000 gallon fuel oil storage tanks with a combined capacity sufficient to operate the diesel at full load for a period of no less than seven days, plus a margin to allow for periodic testing. Fuel oil is transferred by gravity from the storage tanks to the day tank which is located within retaining walls inside the Diesel Building. The day tank capacity is sufficient to operate the diesel generator engine for a minimum of 60 minutes at full load. The fuel oil is pumped from the day tank to the engine by an engine driven fuel oil pump. Excess fuel not consumed by the engine is returned to the day tank via the fuel return line.

NRC FORM 366A (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027			
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1. FACILITY NAME		☒ 050 ☐ 052		2. DOCKET NUMBER 00413, 00414		3. LER NUMBER		YEAR 2025 SEQUENTIAL NUMBER 002 REV NO. 00	
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Technical Specifications: Catawba TS 3.8.1 governs the DGs for each unit that is in Modes 1, 2, 3, and 4. Limiting Condition for Operation (LCO) 3.8.1(b) requires two operable DGs capable of supplying the Onsite Essential Auxiliary Power Systems. With one LCO 3.8.1(b) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action B.6. LCO 3.8.1(d) requires two DGs from the opposite unit necessary to supply power to the shared systems and the Nuclear Service Water System (NSWS) pump(s). With one LCO 3.8.1 (d) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action D.6. If an inoperable DG is not restored to operable status within 72 hours, the unit must be placed in Mode 3 within 6 hours and in Mode 5 within 36 hours per LCO 3.8.1 Required Actions I.1 and I.2. In order to extend the Completion Time for an inoperable DG from 72 hours to 14 days, it is necessary to ensure the availability of the ESPS within 1 hour of entry into TS 3.8.1 LCO and every 12 hours thereafter. EVENT DESCRIPTION On 4/23/2025, Catawba Unit 1 was in Mode 1 operating at approximately 100 percent power. At 0248 hours, the 1B Emergency Diesel Generator (EDG) was started for a 24 hour surveillance. Approximately 11 hours into the surveillance a leak originated on the common fuel supply line under a compression fitting. The leak was estimated to be greater than 0.5 gpm and at 1457 hours the 1B EDG was manually secured and at 1459 hours the 1B EDG was declared inoperable. Repairs were made and the 1B EDG was returned to service on 4/25/2025 at 0600 hours. Prior to the aborted 24 hour surveillance, the 1B diesel successfully performed an operability test on 3/26/2025. An analysis was performed to evaluate past operability for the 1B EDG, specifically reviewing the amount of additional EDG operating time prior to complete fracture of the fuel supply line, fuel oil supply and demand and the potential for the 1B EDG to continue receiving a sufficient rate and volume of fuel with a fractured supply line. This analysis concluded the 1B EDG would have continued to operate with a complete fracture of the fuel line; however, the fuel storage available to support the seven day mission time of the 1B EDG would be challenged. This analysis was finalized on 6/13/2025, starting the 60-day reportability clock for this LER. This event is reportable under 10 CFR 50.73(a)(2)(i)(B) as any operation or condition which was prohibited by the plant's Technical Specifications. Forensics determined a crack originated some time ago on the fuel supply line and propagated during the attempted 24 hour surveillance, resulting in a fuel leak that would have challenged the seven day mission time for the 1B EDG. This condition existed for a time period exceeding the 14-day allowed outage time for an inoperable EDG.									



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CAUSAL FACTORS

The affected section of tubing was inspected and subsequently removed and sent for metallurgical analysis. The leak was caused by an outer diameter initiated through wall circumferential crack that extended approximately 300 degrees around the tube circumference within a compression fitting. The failure was attributed to high-cycle fatigue, consistent with mechanical vibration occurring during engine operation. Beach marks and multiple arced crack arrest lines on the fracture surface that emanated from the origin are indicative of multiple periods of active crack growth aligning with periodic operation of the diesel. Oxidation along the beach marks at the origin suggest the crack initiated some time ago. While the investigation identified the crack initiated some time ago, the location was hidden within the compression fitting, was not through wall prior to the surveillance, and there was no previous indication of leakage. There is no history of maintenance at the fitting where the crack was located and there was no operating history of leaks or cracks at this location on any emergency diesel generator at Catawba.

The tubing configuration on the 1B EDG introduced additional mechanical stress at the tubing to after-cooler wall interface, creating a rubbing condition. While it is noted the leak was not at the location of the rubbing, it is possible the vibration due to the tubing contact with the aftercooler wall was translated to the crack location.

An additional contributing factor to the crack was a loose horizontal clamp in close proximity to the fitting where the crack was located. The exact impact of the clamp being loose cannot be determined nor how long the clamp was loose. Catawba currently performs a 6-year preventative maintenance task to replace missing or broken clamps as necessary. The last completion of this activity for the 1B EDG was on 9/2/2020, with no issues noted. Additionally, there are no specific preventative maintenance requirements for clamp inspection or replacement per the Enterprise Preventative Program Manual.

CORRECTIVE ACTIONS

Immediate:

1. Remove and replace affected section of fuel oil tubing and perform metallurgical analysis on leak location.
2. Remove and replace similar location of fuel oil tubing for the opposite train (1A) EDG and perform metallurgical analysis.

Planned:

1. Reroute the 1B EDG fuel oil tubing to eliminate contact from the side of the aftercooler wall.
2. Revise preventative maintenance tasks to add additional instructions and criteria.
3. Remove and replace similar locations of fuel oil tubing for the 2A and 2B EDG and perform metallurgical analysis.

SAFETY ANALYSIS

The fuel leak of the 1B Emergency Diesel Generator presents a reduction in the reliability and availability of the Catawba Unit 1 emergency power system. There was no plant transient, radiological release, or other challenge to normal plant operation to cause any impacts to the public health and safety.



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Catawba Unit 1 has a considerable level of defense-in-depth besides having two DG trains. The ESPS provides a means to restore AC power to the 1B Essential Bus (1ETB) if the 1B DG failed. Similarly, another recovery action is the ability to align offsite power from Unit 2 if available via the Shared Auxiliary Transformer SATB. The Safe-Shutdown Facility (SSF) also provides an alternative means of maintaining safe shutdown conditions with the SSF Diesel Generator powering the Standby Makeup Pump for reactor coolant pump (NCP) seal cooling and instrumentation and controls for operation of Turbine-Driven Auxiliary Feedwater Pump. The final level of safety is the ability to use FLEX equipment and strategies to provide alternative core cooling to prevent core damage.

With these factors taken together there was no impact to the health and safety of the public.

ADDITIONAL INFORMATION

There have been no previous Licensee Event Reports at Catawba Nuclear Station in the last three years with the same causal factor as this issue.